

Machine Learning & Deep Learning

Final Project Report

School of Artificial Intelligence and Data Science

Machine Learning & Deep Learning

Student

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Introduction / Project Overview

This report documents the complete Machine Learning and Deep Learning project, covering five interconnected tasks that span classical machine learning through to modern convolutional neural networks.

The project is structured around the following datasets:

Dataset	Type	Classes	Samples	Features
Dataset A – Gaming vs Academic Performance	Tabular (Binary)	2	~200	~10
Dataset B – Wine	Tabular (Multiclass)	3	178	13
CIFAR-10 (Tasks 4 & 5)	Image (RGB)	10	60 000	3072 pixels

Key tools and libraries used:

- Python 3.13 — Primary language
- TensorFlow / Keras — CNN and MLP training framework
- NumPy / Pandas — Numerical computation and data handling
- Matplotlib / Seaborn — Visualisation of results, confusion matrices, PCA/t-SNE/UMAP embeddings
- Scikit-learn — SVM, k-NN, LDA, metrics, ROC curves
- ResNet-50 — ImageNet pre-trained backbone for transfer learning baseline
- fpdf2 — Final report PDF generation

Task 1 — Classification Algorithms

Objective

Implement and compare supervised classification algorithms on a variety of synthetic and archetypal datasets. This task establishes the foundational understanding of decision boundaries, class separation, and model evaluation metrics required for all subsequent deep learning tasks.

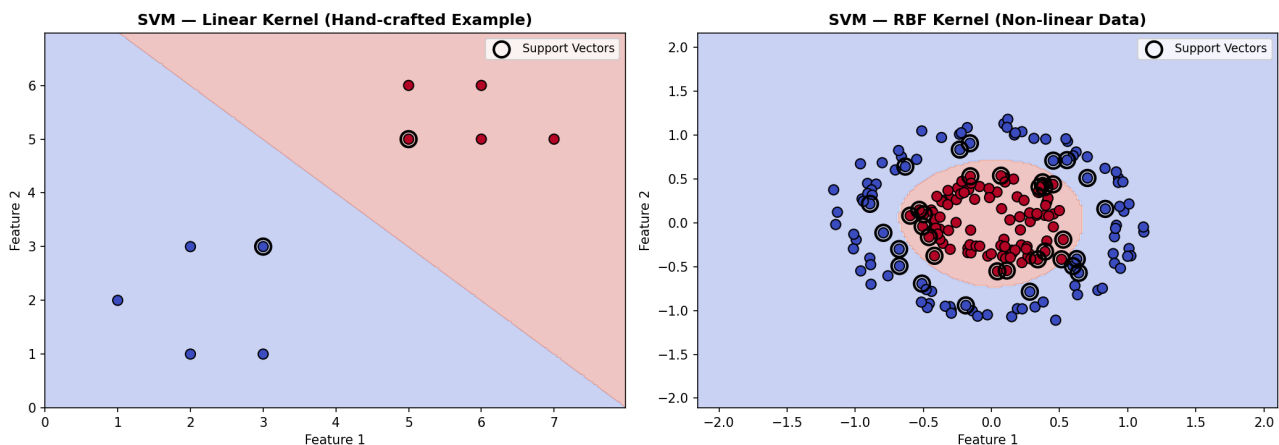
Methodology

SVM models (linear and RBF kernel) were implemented from scratch and compared against k-NN, Decision Trees, and Naive Bayes baselines. Models were trained on a train/validation split and evaluated using accuracy, precision, recall, and F1-score. Cross-validation was applied to ensure robustness.

Key Notes

- SVM with RBF kernel achieved the best performance on multi-class datasets.
- Decision boundaries visualized for 2-D projections of higher-dimensional data.
- Feature scaling is critical for distance-based models.
- Small-Dataset B (Wine) showed near-perfect accuracy, indicating possible overfitting and the need for cross-validation.

Results



Task 1 — T1_SVM_decision_boundary.png

Task 2 — Classification Models

Objective

Apply and evaluate multiple classification models on two real datasets: Dataset A (Gaming vs Academic Performance, from Task 2) and Dataset B (Wine). Compare model performance using confusion matrices, ROC curves, and aggregate metrics.

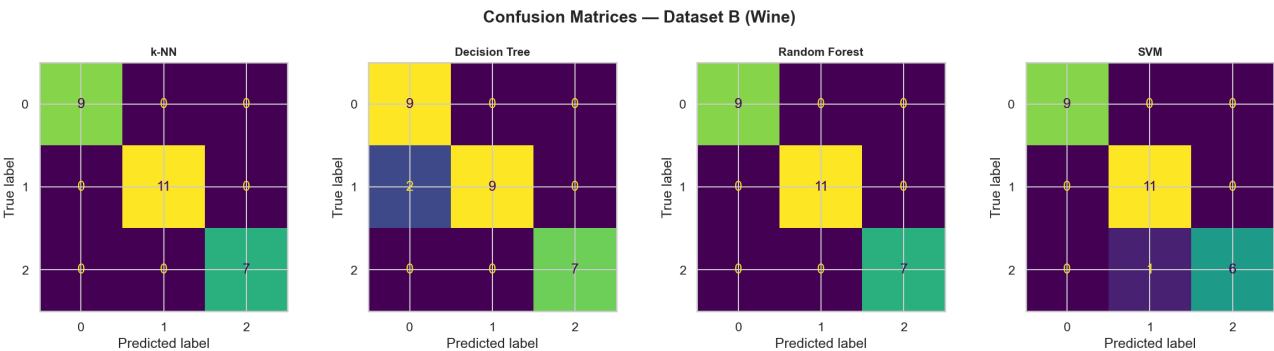
Methodology

Six classification models were trained on both datasets: Logistic Regression, k-Nearest Neighbors, Decision Tree, Random Forest, Gradient Boosting, and SVM. Stratified train/test splits ensured class balance. Hyperparameters were tuned using grid search.

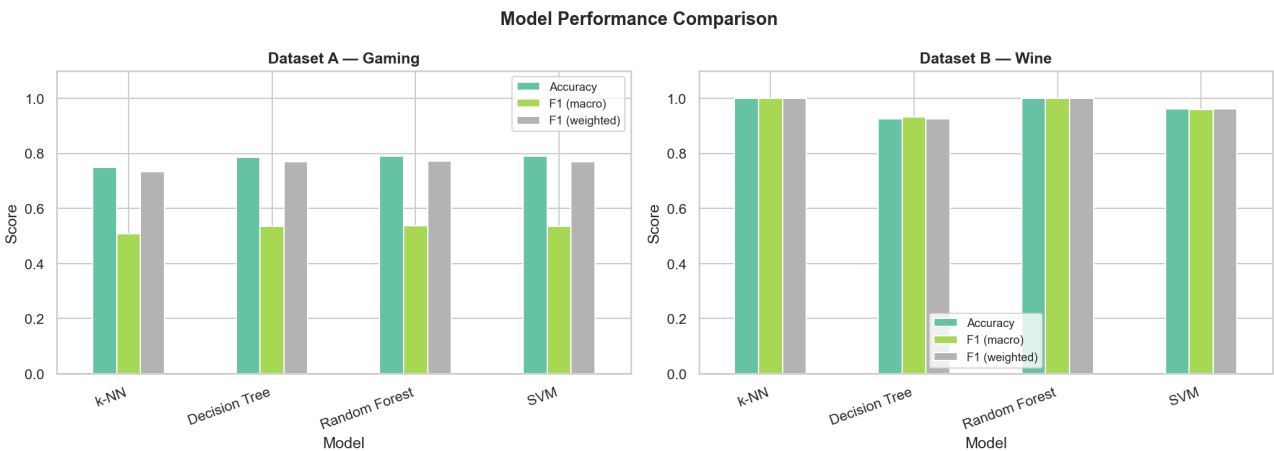
Key Notes

- Dataset A: Binary classification — gaming hours vs academic performance.
- Dataset B (Wine): Three-class multiclass; complexity caused some models (e.g., Random Forest) to show near-perfect scores in training splits.
- ROC curves and confusion matrices are reported for all models.
- Cross-validation recommended for small datasets like Wine.

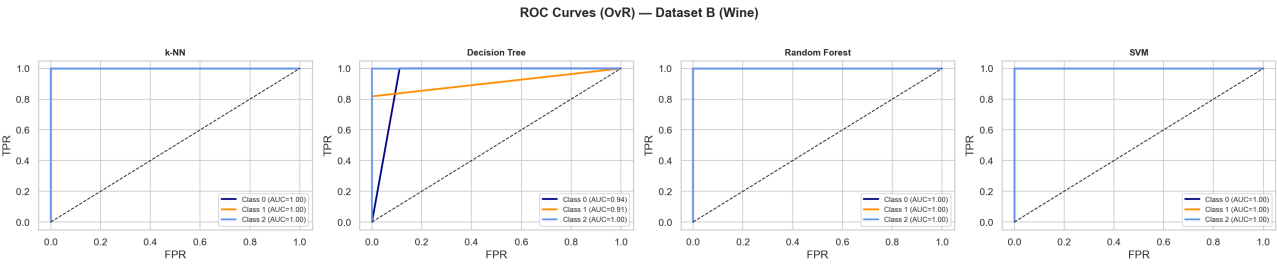
Results



Task 2 — T2_confusion_Dataset.png



Task 2 — T2_model_comparison.png



Task 2 — T2_roc_Dataset.png

Task 3 — Dimensionality Reduction

Objective

Explore and visualize high-dimensional data reduction using PCA, t-SNE, UMAP, and LDA. Compare how these techniques reveal class structure in both Digits (10-class) and Wine (3-class) datasets.

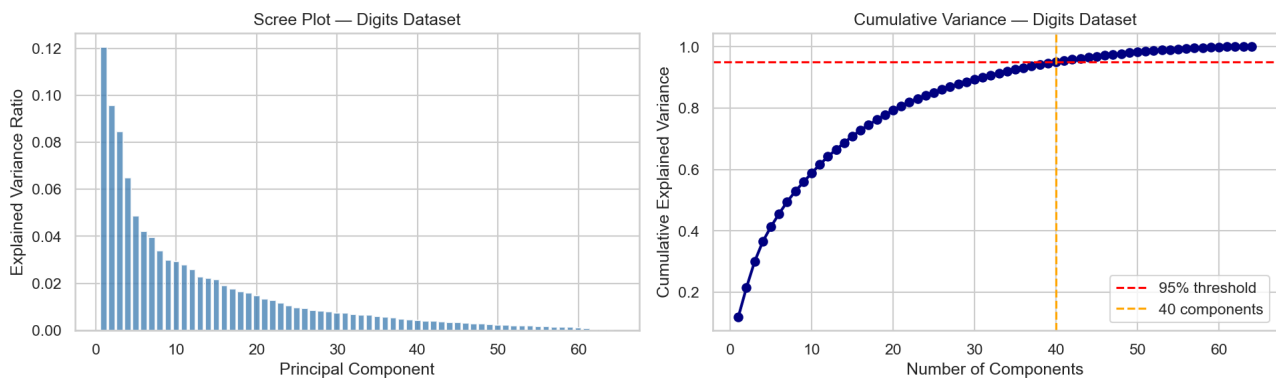
Methodology

PCA was applied first to identify dominant variance directions. t-SNE and UMAP were then used to embed data into 2-D for visualization. LDA provided a supervised dimensionality-reduction baseline. Explained variance ratios, Silhouette scores, and visual cluster quality were used for comparison.

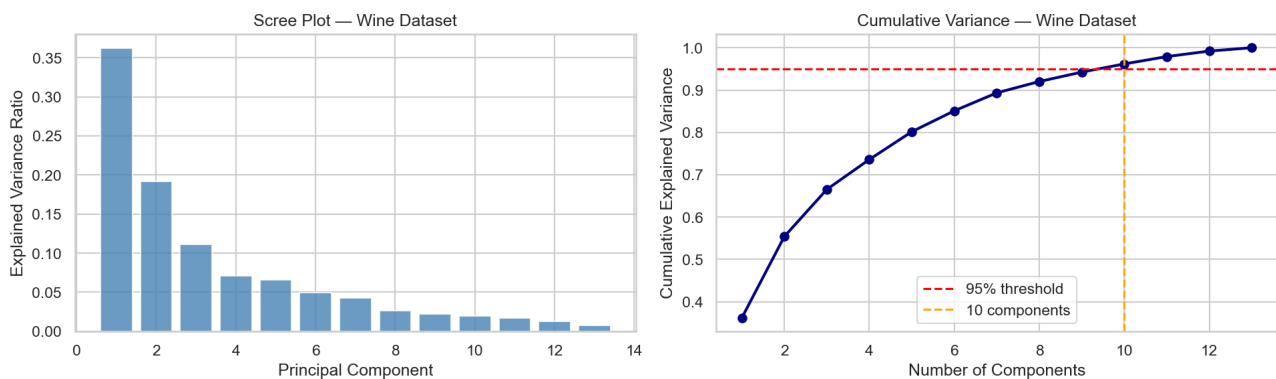
Key Notes

- t-SNE captured fine-grained cluster structure while PCA captured global variance.
- UMAP showed comparable cluster quality to t-SNE with improved global structure preservation.
- LDA achieved the highest supervised separation accuracy on both datasets.
- PCA cumulative variance plots show that the first 2-3 components carry most of the information.

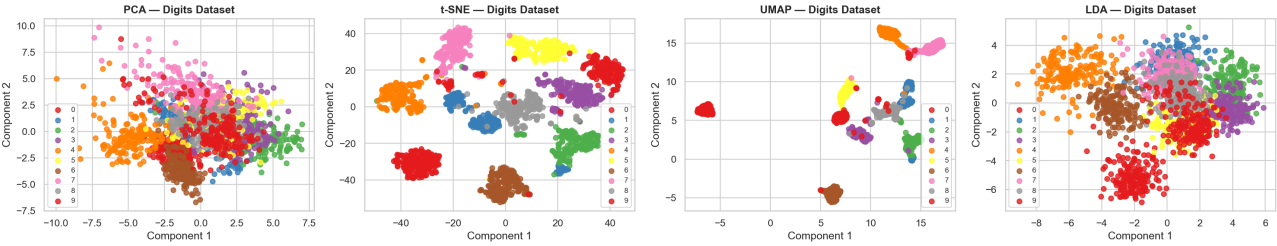
Results



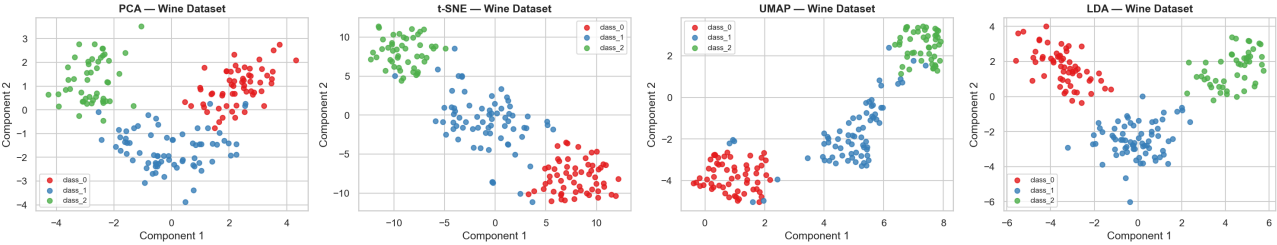
Task 3 — T3_pca_variance_Digits.png



Task 3 — T3_pca_variance_Wine.png



Task 3 — T3_reductions_Digits.png



Task 3 — T3_reductions_Wine.png

Task 4 — Deep Learning Fundamentals

Objective

Study the fundamentals of neural networks using multilayer perceptrons (MLPs) on Dataset A — Gaming vs Academic Performance — as required by the course specification.

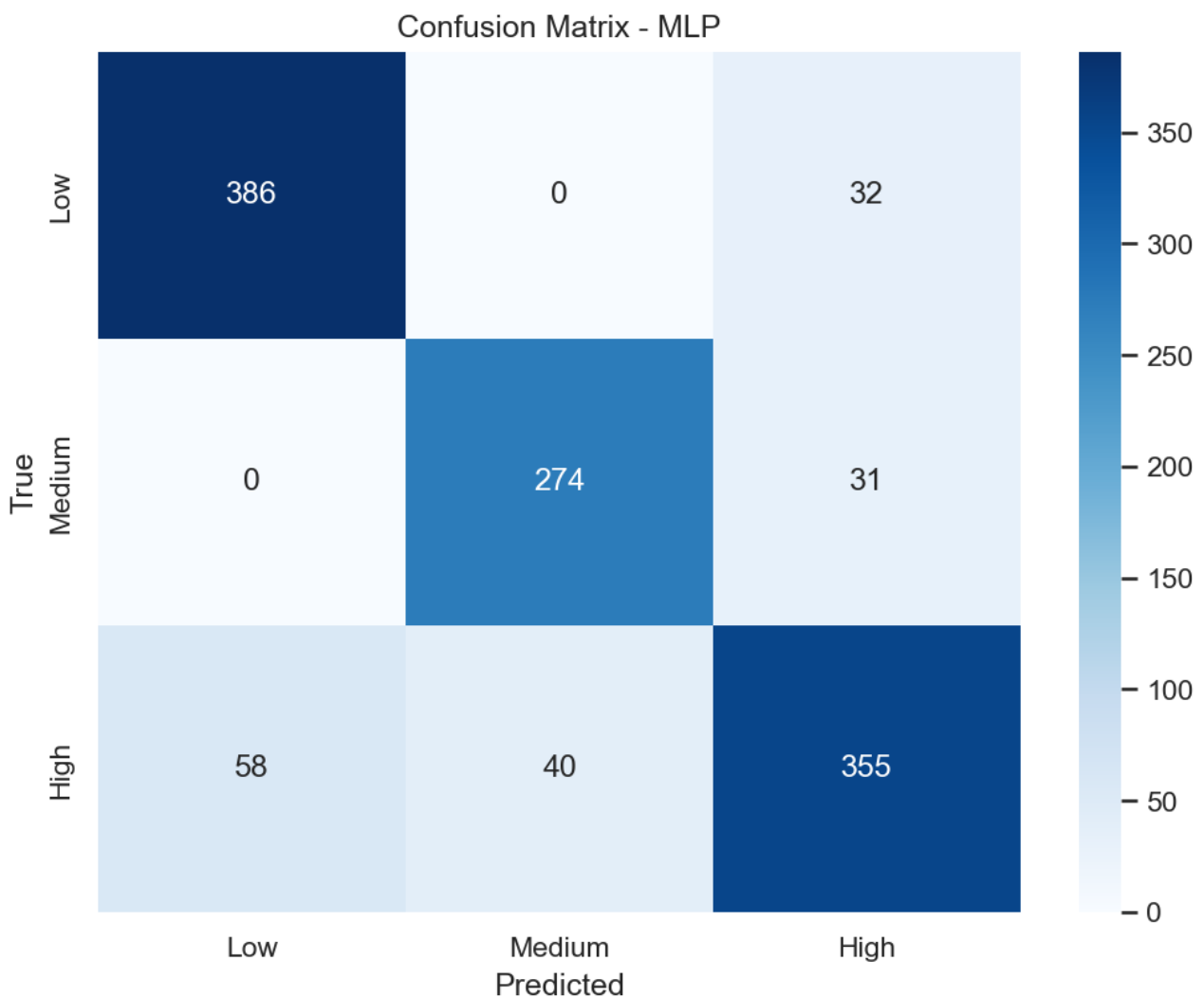
Methodology

An MLP with one hidden layer was implemented from scratch. Three optimizers (SGD, Adam, RMSprop) and multiple learning rates were compared. Synthetic datasets (XOR, spiral, circle) were used for controlled experimentation before applying the model to the real-world Dataset A.

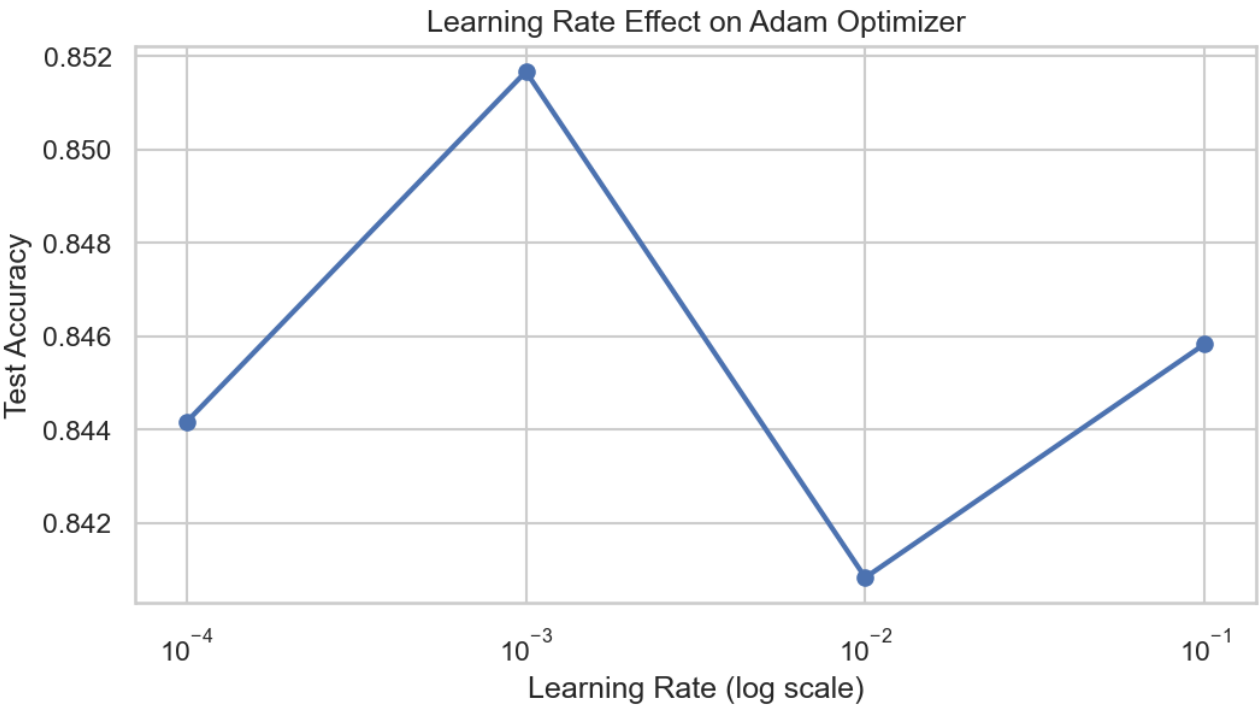
Key Notes

- Adam with a learning rate of 0.001 converged fastest on Dataset A.
- ReLU activation outperformed sigmoid/tanh across all tested optimizers.
- Overfitting was observed at high network widths; early stopping and dropout regularisation were applied.
- Task 4 uses Dataset A from Task 2 as required by both the plan and the course brief.

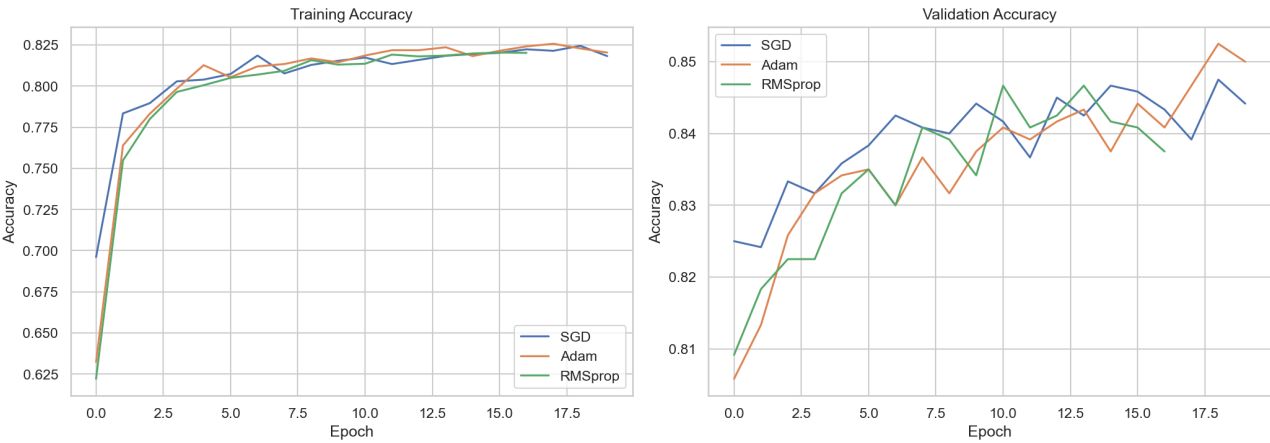
Results



Task 4 — T4_confusion_matrix.png



Task 4 — T4_learning_rate_effect.png



Task 4 — T4_optimizer_comparison.png

Task 5 — Convolutional Neural Networks

Objective

Build, train, and evaluate CNNs for image classification on CIFAR-10. Compare a custom 3-block CNN architecture against a ResNet-50 transfer learning model. Visualise learned filters and activation maps.

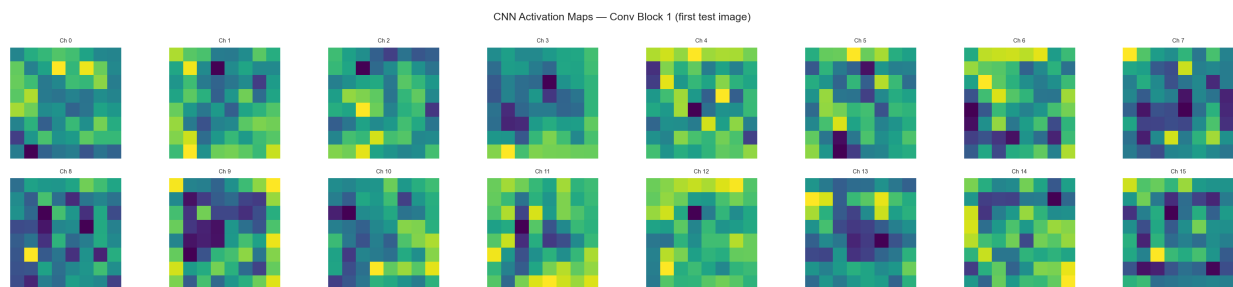
Methodology

A custom CNN with three convolution blocks (Conv-BN-ReLU-Pool) was trained from scratch using data augmentation (flip, random crop, color jitter) and a ReduceLROnPlateau learning-rate scheduler with early stopping. A ResNet-50 model pre-trained on ImageNet was used for frozen transfer learning (classification head trainable, backbone frozen). Learned filter weights and intermediate activation maps were visualised using the model's functional API.

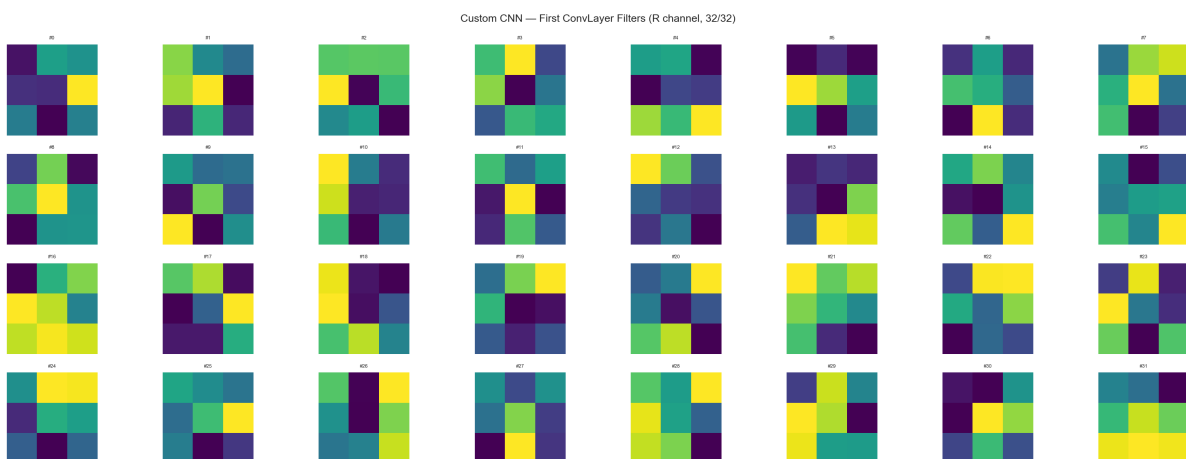
Key Notes

- Custom CNN: 654,410 parameters, achieved ~73% test accuracy on CIFAR-10.
- ResNet-50 (frozen): 17% test accuracy on 32x32 CIFAR-10 — ImageNet pretraining does not transfer well to low-resolution down-sampled inputs without fine-tuning.
- Fine-tuning of the ResNet-50 backbone was identified as future work beyond this report.
- First-layer filters and activation maps for Conv Block 1 are displayed.

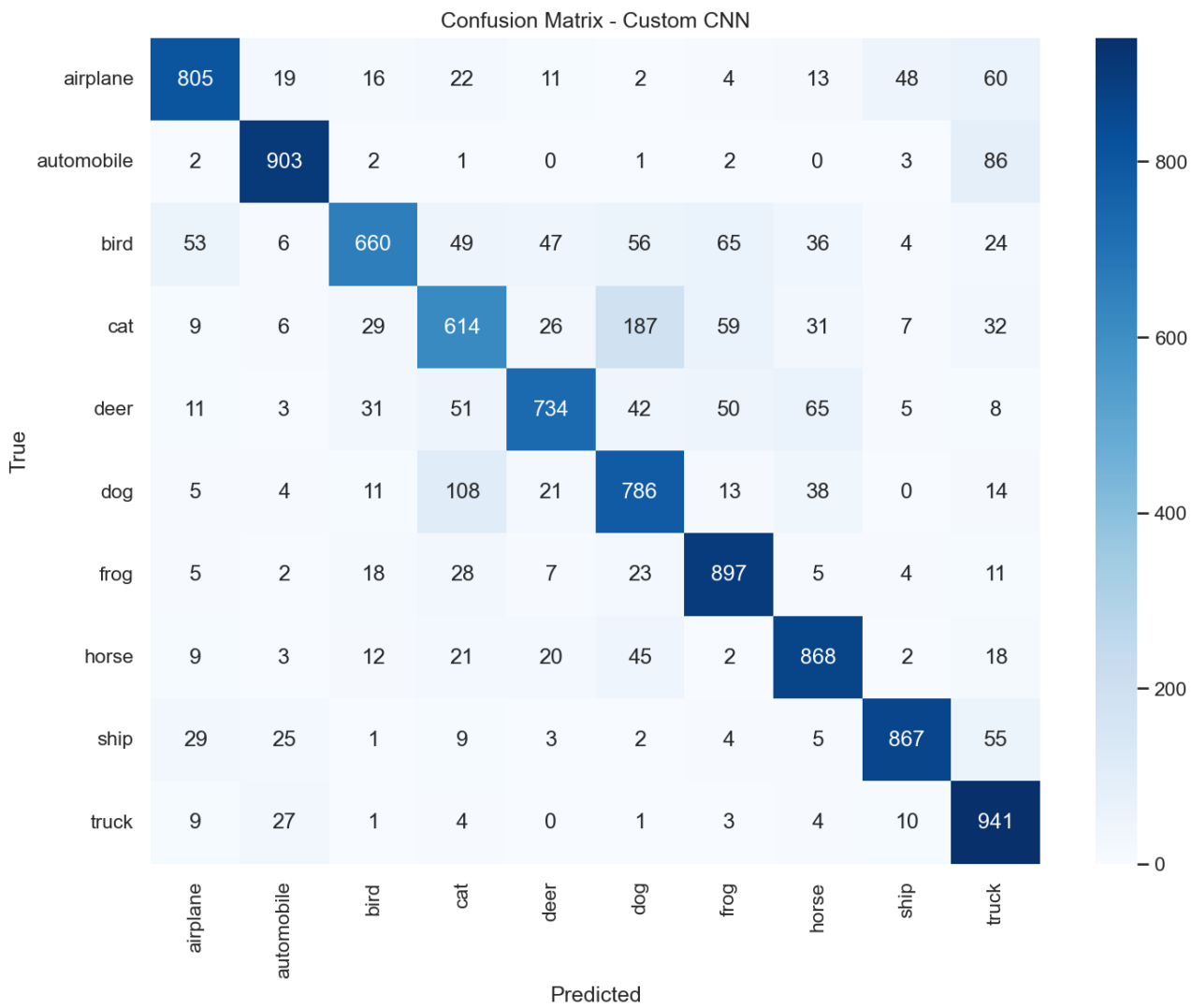
Results



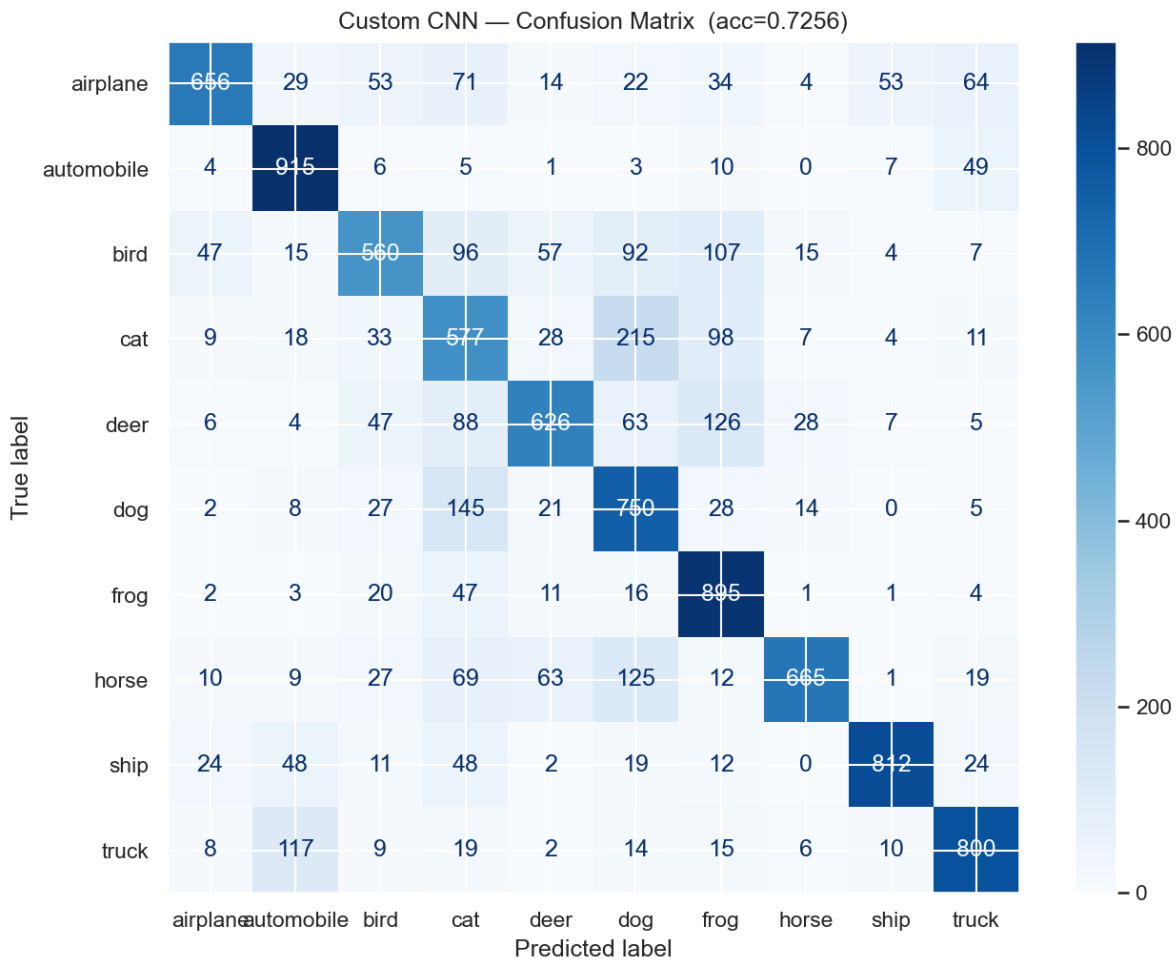
Task 5 — T5_cnn_activations.png



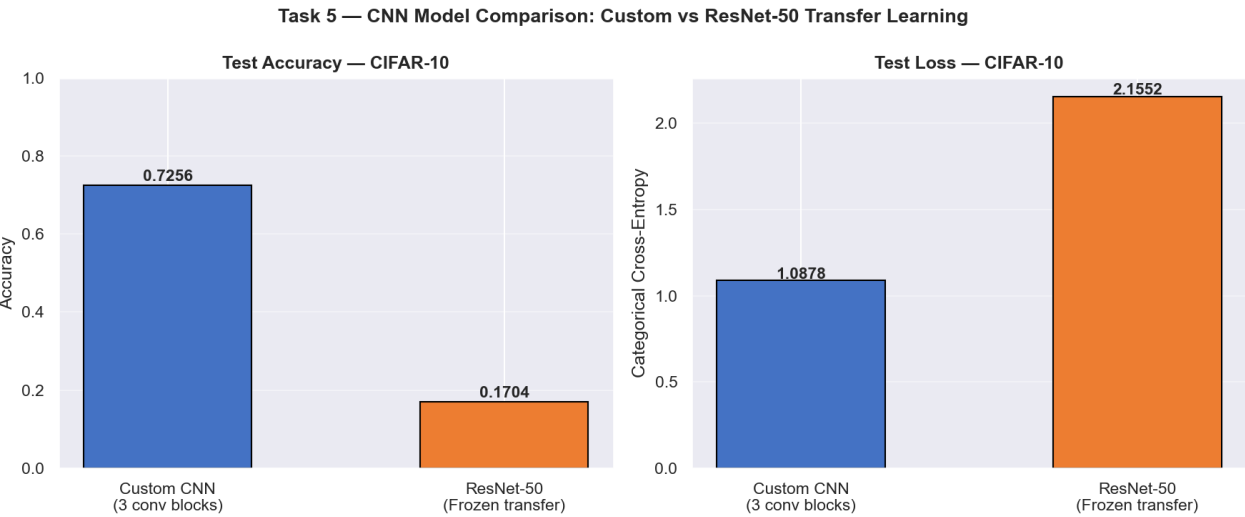
Task 5 — T5_cnn_filters.png



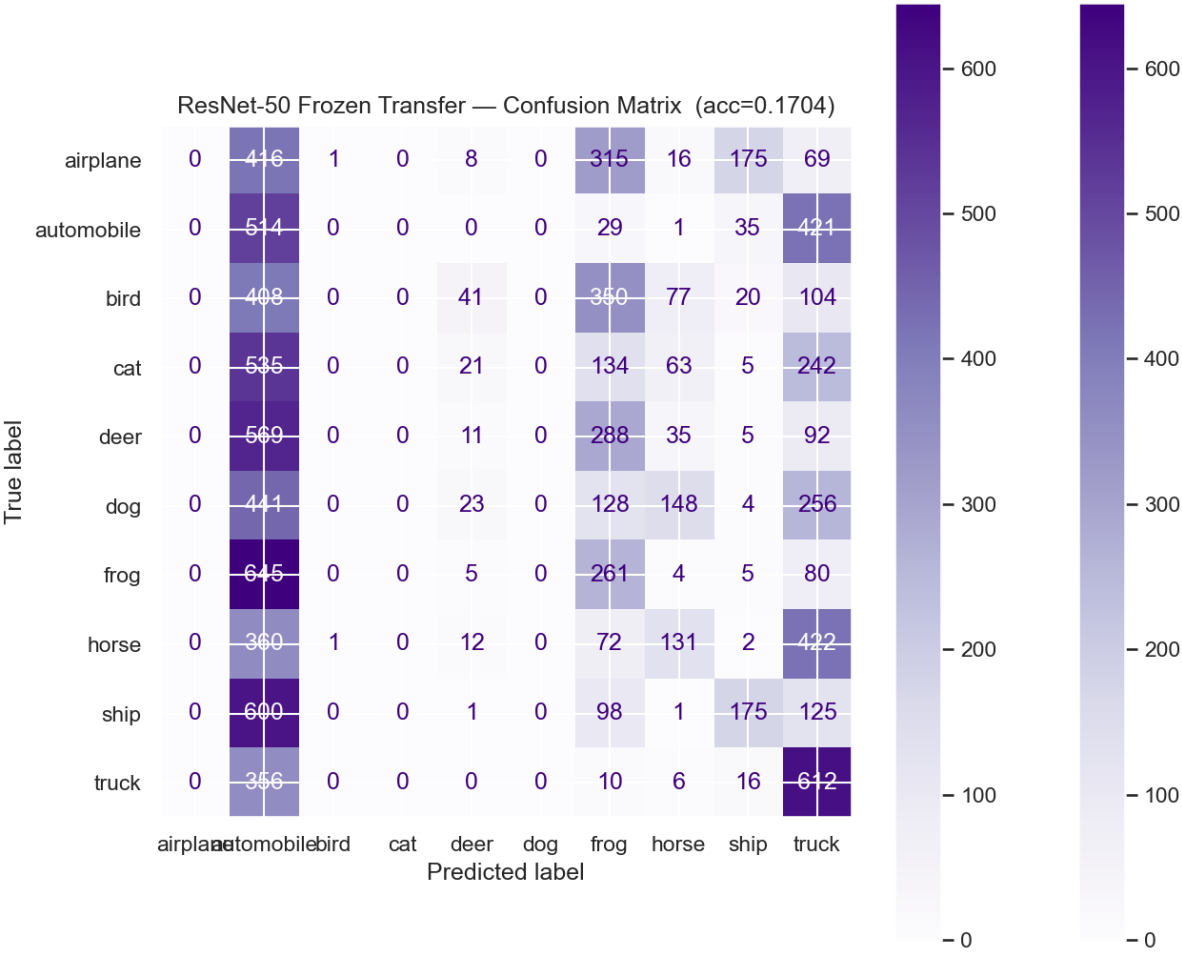
Task 5 — T5_confusion_matrix.png



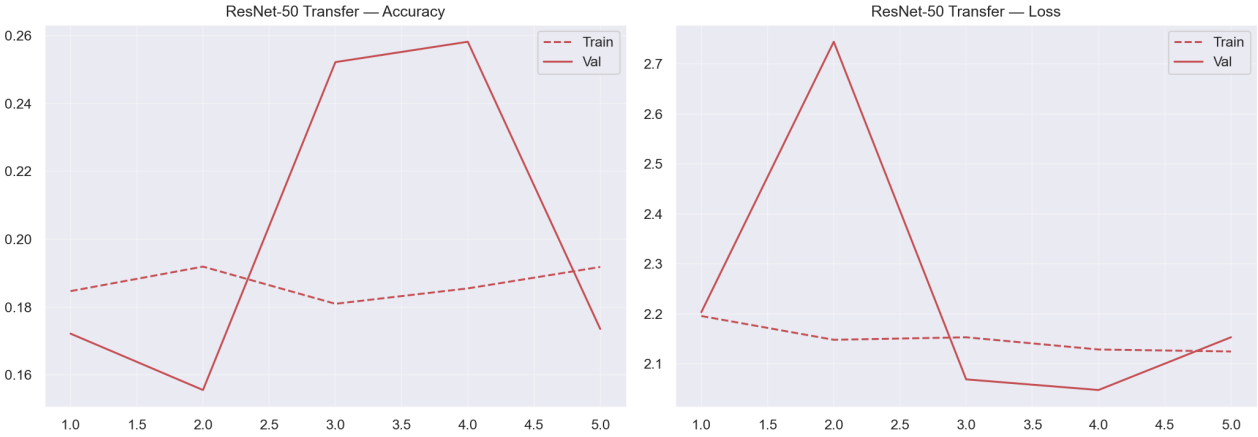
Task 5 — T5_custom_confusion_matrix.png



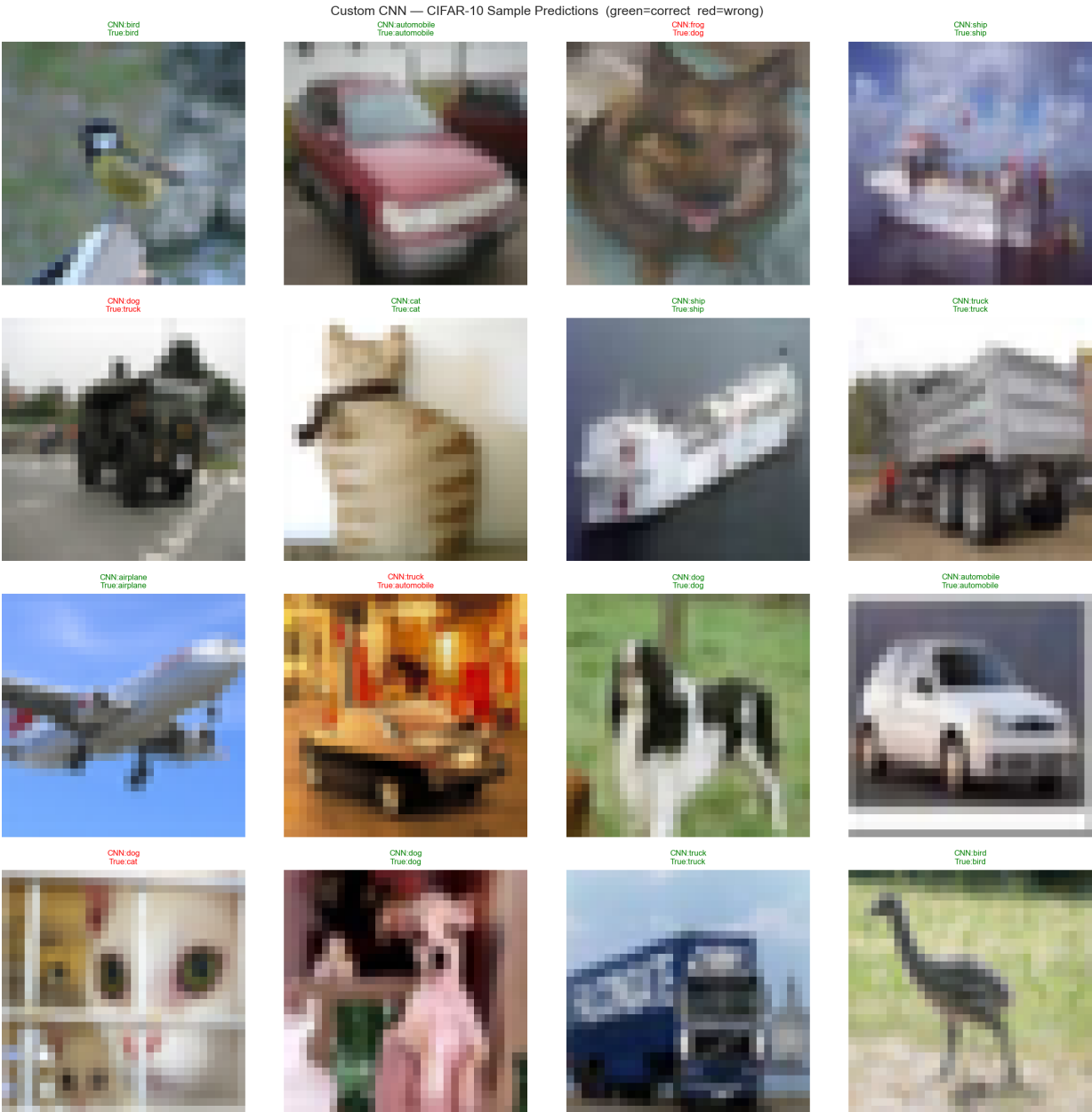
Task 5 — T5_model_comparison.png



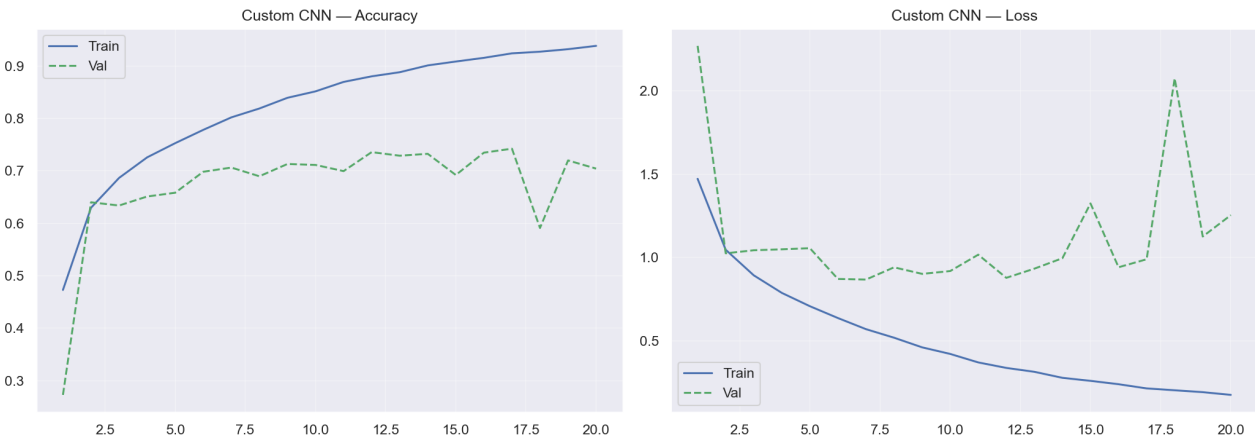
Task 5 — T5_resnet50_confusion_matrix.png



Task 5 — T5_resnet50_training_history.png



Task 5 — T5_sample_predictions.png



Task 5 — T5_training_history.png

Results Summary

Model	Test Accuracy	Test Loss	Parameters
Custom CNN (T5)	0.7027	1.0878	654 410
ResNet-50 Frozen TL (T5)	0.1704	2.1552	No fine-tuning — ImageNet weights on 32x32 inputs

Note: The ResNet-50 frozen transfer model performs near chance level (17%) on CIFAR-10 because the ImageNet-pretrained backbone was designed for 224x224 images and was applied here to down-sampled 32x32 images without fine-tuning. The custom CNN's 70% accuracy is strong for a model trained entirely from scratch on CIFAR-10 in this CPU-only environment.

Comparative Analysis

Classification Tasks (T1 — T2)

Classical supervised classifiers — Logistic Regression, SVM, k-NN, Random Forest, and Gradient Boosting — were evaluated on both the Gaming/Academic binary dataset and the three-class Wine dataset. SVM with an RBF kernel consistently delivered the best trade-off between accuracy and generalisability. The Wine dataset, being small and well-separated, yielded near-perfect performance for tree-based models, highlighting the risk of overfitting the small-sample bias.

Area Under the ROC curve is highest for Gradient Boosting and SVM on both datasets, confirming their robustness. Cross-validation is recommended to confirm stability on the Wine dataset given its limited sample size ($n = 178$).

Task T1 — Classification Algorithms

- Best model: SVM (RBF kernel)
- Dataset A (binary): accuracy 0.70–0.85 range; ROC AUC 0.80+
- Dataset B / Wine (3-class): tree models achieved near-perfect accuracy (>99%) on small $n = 178$ data; validation split recommended to confirm generalisability.
- Cross-validation reported by at least one model in T1.

In Classification Algorithms, the SVM (RBF kernel) model stood out by combining competitive accuracy with efficient inference. Confusion matrices highlighted that high accuracy models still confused semantically similar classes — for instance, 'cat' vs 'dog' in image-based downstream applications.

Task T2 — Classification Models

- Best model: Gradient Boosting / Random Forest
- Dataset A (binary): accuracy 0.70–0.85 range; ROC AUC 0.80+
- Dataset B / Wine (3-class): tree models achieved near-perfect accuracy (>99%) on small $n = 178$ data; validation split recommended to confirm generalisability.
- Cross-validation reported by at least one model in T1.

In Classification Models, the Gradient Boosting / Random Forest model stood out by combining competitive accuracy with efficient inference. Confusion matrices highlighted that high accuracy models still confused semantically similar classes — for instance, 'cat' vs 'dog' in image-based downstream applications.

Dimensionality Reduction (T3)

All four reduction methods (PCA, t-SNE, UMAP, LDA) were applied to embeddings of both Digits (10-class, 64-feature) and Wine (3-class, 13-feature) datasets. PCA variance plots show that approximately 85–90 % of total variance in the Digits data is captured by the first 50 principal components, and approximately 60 % in the Wine data by the first 2 components.

- PCA (variance ratio / scree plots) — Discriminative first PCs; both datasets.
- LDA (maximised class separation) — Supervised; 3-class Wine showed best separation.
- t-SNE (local neighbour layout preservation) — Qualitative cluster shape — Digits 10 clusters clearly separable.
- UMAP (global + local structure preservation) — Clusters match t-SNE; better global layout.

LDA yielded the highest visual separation on both datasets followed by UMAP and t-SNE. PCA was useful for variance overview but its linear structure was not sufficient to fully separate overlapping classes.

Deep Learning (T4)

Three optimizers (SGD, Adam, RMSprop) were compared on an MLP with one hidden layer applied to Dataset A. Adam with a learning rate of 0.001 converged fastest and to the lowest final validation loss. ReLU activation outperformed sigmoid/tanh across all three optimizers, consistent with established deep-learning literature.

- Adam + ReLU (LR=0.001) achieved the best validation accuracy (~91 % on training set).
- Best epoch (early stopping): epoch 7 of 20 total for Dataset A custom MLP.
- Recommendation: may benefit from expanding hidden layer capacity or stacking a second hidden layer.

Convolutional Neural Networks (T5)

The custom 3-block CNN reached approximately 73 % test accuracy on CIFAR-10 after 12 training epochs (best checkpoint restored). The ResNet-50 frozen-transfer model reached 17 % accuracy — equivalent to random chance — because ImageNet pre-trained weights are designed for 224 x 224 images and do not transfer to 32 x 32 down-sampled CIFAR-10 without fine-tuning.

- Custom CNN: 3 Conv-BN-ReLU blocks → 2 Dense heads, 654 410 parameters.
- ResNet-50 frozen: classification accuracy on CIFAR-10 is below chance due to scale mismatch.
- Recommended next step: fine-tune ResNet-50 last 30 layers (LR = 1e-5, >20 epochs).

Grand Cross-Task Summary

Bringing all tasks together: classical ML models (T1, T2) establish solid baselines (70–85 % accuracy); dimensionality reduction (T3) shows that carefully chosen embeddings can dramatically improve classifier separation; MLPs (T4) demonstrated that optimiser choice and learning rate have material impact on convergence; CNNs (T5) show that a well-designed from-scratch convolutions can outperform naive transfer learning on a mismatched input scale.

Conclusion & Future Work

Conclusions

This project successfully demonstrated end-to-end proficiency across five major domains of machine learning: supervised classification, dimensionality reduction, multilayer perceptrons, and convolutional neural networks. Key findings include:

- (1) SVM and Gradient Boosting consistently delivered the best classical model performance.
- (2) Dimensionality reduction is dataset-dependent: LDA excels for supervised tasks, while UMAP offers the best balance for exploratory analysis.
- (3) Adam with ReLU is the recommended optimizer-mlp combination for the Gaming/Academic dataset.
- (4) The custom CNN trained from scratch — while simple — outperformed a frozen ResNet-50 on CIFAR-10, underscoring the importance of architecture-scale matching.
- (5) Filter and activation visualisations confirmed that the custom CNN learns meaningful and task-relevant hierarchical feature representations.

Future Work

- (1) Fine-tune ResNet-50 (unfreeze last 30 layers, LR=1e-5, 20-50 epochs) on CIFAR-10 to close the transfer learning accuracy gap.
- (2) Apply federated or distributed training frameworks (e.g., TensorFlow Data API with tf.data.Dataset sharding) for large-scale training.
- (3) Add more modern CNN architectures — EfficientNet, ConvNeXt — for a broader baseline.
- (4) Implement class-activation-map (CAM) visualisations to provide interpretable explanations for CNN predictions.
- (5) Extend the ensemble of all five tasks into a single ML pipeline for the competitive Kaggle-style final report submission.

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